Dataset Details

Source: **Crop Yield Data with Soil and Weather Dataset** (Kaggle, CC0) - https://www.kaggle.com/datasets/anshumish/crop-yield-data-with-soil-and-weather-dataset

Row counts

Item	Value
Source rows (crop_yield.csv)	19,689
Dropped: missing / invalid / exact duplicates / duplicate keys	0 / 0 / 0 / 0
Valid REAL rows	19,689
Labelled SYNTHETIC rows added	35,311
Target rows (capstone brief)	55,000
Final working rows	55,000
Target met with real data alone?	No
Strategy	all real rows used + labelled synthetic augmentation
Random seed (used only if sampling were needed)	42
Dataset version (hash of cases.csv)	329b6c309935

How 55,000 records are reached: the published dataset contains only 19,689 valid real records. At the project owner's request the working dataset is topped up to 55,000 with **35,311 labelled synthetic records** (is_synthetic = true, case ids start 'TS-', source_row negative). They are NOT historical observations. Method: Each synthetic row perturbs a randomly chosen real row of the same crop/state/season (year drawn from the group's observed range; real state soil and state-year weather re-attached); area and per-hectare inputs get log-normal noise (sigma 0.20), yield sigma 0.15, production = yield x area. No new crop-state pairs are created. Safeguards: synthetic rows are badged in the UI, can be excluded with a filter / toggle, and are never used for yield-model training or testing, statistics, impact analysis or suitability envelopes (those use the 19,689 real rows only). Re-run scripts/prepare_dataset.py --no-augment for a real-only dataset.

Schema and field meanings

Field	Meaning / units	Origin
crop, year, season, state	Crop name, calendar year, growing season (Kharif, Rabi, Summer, Winter, Autumn, Whole Year), Indian state	crop_yield.csv
area	Cultivated area (hectares)	crop_yield.csv
production	Production (crop-specific units)	crop_yield.csv
yield	Reported yield, ~ production/area. yield ~ production / area (median ratio of reported yield to production/area = 1.000). Units are NOT stated by the source; they are crop-specific (typically tonnes/hectare, but e.g. coconut is reported per-nut). Yields are therefore only comparable within a crop.	crop_yield.csv
fertilizer, pesticide	Record totals (dataset units, not per hectare)	crop_yield.csv
n, p, k, ph	N, P, K, pH are state-level constants (one value per state); units of N/P/K are not stated by the source.	state_soil_data.csv (joined on state)
temperature, rainfall, humidity	temperature (C), rainfall (mm, annual total) and humidity (%) are state-year aggregates, not farm-level.	state_weather_data (joined on state+year)
case_id	TM- + first 10 hex chars of SHA-1(crop|year|season|state), unique per natural key; source_row = 0-based row of crop_yield.csv	derived
fertilizer_per_ha, pesticide_per_ha	total / area	derived
yield_band, ph_class, rainfall_class, temp_class, humidity_class	Descriptive classes with documented thresholds (see below)	derived

Field	Meaning / units	Origin
completeness, flags	Share of soil/weather fields present; data-quality flags	derived

Preprocessing decisions

• Trimmed/collapsed whitespace in crop, season, state labels (season values are space-padded in the source).

• Dropped 0 rows with missing required values and 0 rows with non-positive area or negative production/yield/inputs.

• Removed 0 exact duplicate rows and 0 duplicate (crop, year, season, state) keys (first kept).

• Joined state-level soil N, P, K, pH on state. These are ONE constant value per state, not per-farm measurements.

• Joined state-year weather (annual mean temperature C, total annual rainfall mm, mean humidity %) on (state, year).

• Valid rows available: 19689. Target: 55000. Target NOT reachable with real data; labelled synthetic rows are added (see AUGMENTATION).

• AUGMENTATION: 35311 SYNTHETIC rows were generated to reach 55000 (is_synthetic=True, ids start 'TS-', source_row=-1). They are not historical observations. Each synthetic row perturbs a randomly chosen real row of the same crop/state/season (year drawn from the group's observed range; real state soil and state-year weather re-attached); area and per-hectare inputs get log-normal noise (sigma 0.20), yield sigma 0.15, production = yield x area. No new crop-state pairs are created.

Derived column definitions

• **fertilizer_per_ha**: fertilizer / area

• **pesticide_per_ha**: pesticide / area

• **yield_band**: per-crop percentile of yield: <=25th low, >75th high, else typical

• **ph_class**: bins <5.5 strongly acidic, 5.5-6.5 acidic, 6.5-7.5 neutral, >7.5 alkaline

• **rainfall_class**: bins <800, 800-1500, 1500-2500, >2500 mm

• **temp_class**: bins <20, 20-26, >26 C

• **humidity_class**: bins <55, 55-72, >72 %

• **completeness**: share of the 7 soil/weather fields present

Data-quality findings

• Missing values after merge: none in any column (soil and weather joined for all 55,000 rows; synthetic rows reuse the real state soil and state-year weather).

• Flags: 333 records with zero production (kept, flagged: could be crop failure or unreported); 6420 records (12%) where reported yield differs from production/area by more than 10% (kept, flagged).

• Extreme values (> Q3 + 3xIQR, inspected not deleted): {'yield': 6379, 'area': 6503, 'fertilizer': 6473, 'pesticide': 6484}. These are legitimate large-area / high-magnitude observations (e.g. coconut yields are per nut).

• Coverage: 55 crops, 30 states, years 1997-2020 (2020 is partial: 37 rows).

• Not available in the source and therefore never inferred: soil type, irrigation, farm location, crop variety.

Limitations

• Soil values are state constants (30 distinct profiles): within a state, soil cannot explain yield differences, and soil features act partly as state identifiers.

• Weather is a state-year aggregate and does not describe an individual farm.

- Yield units are crop-specific and not stated by the source; cross-crop yield comparison is invalid.
- Fertilizer and pesticide are totals that scale with area; per-hectare values are derived.